Online EWC Savings

Constant-memory consolidation vs regular per-task EWC storage

99%

lower EWC state at 100 tasks

O(T) -> O(1)

50x

fewer total penalty terms at 100 tasks

Regular EWC grows with every past task. Online EWC keeps a single consolidated Fisher + parameter snapshot.

For the Figure A MLP, the absolute memory is small - the key advantage is scalability.

Online EWC savings model - Figure A MLP: 478,410 parameters, float32

What exactly is being saved?

The saving is not in the neural network itself - it is in the extra EWC state kept between tasks.

478,410

trainable parameters in the Figure A MLP

1.83 MiB

one float32 array with this parameter count

3.65 MiB

one Fisher + one parameter snapshot

Storage model

Regular EWC state = 2 x (tasks - 1) x parameters x 4 bytes

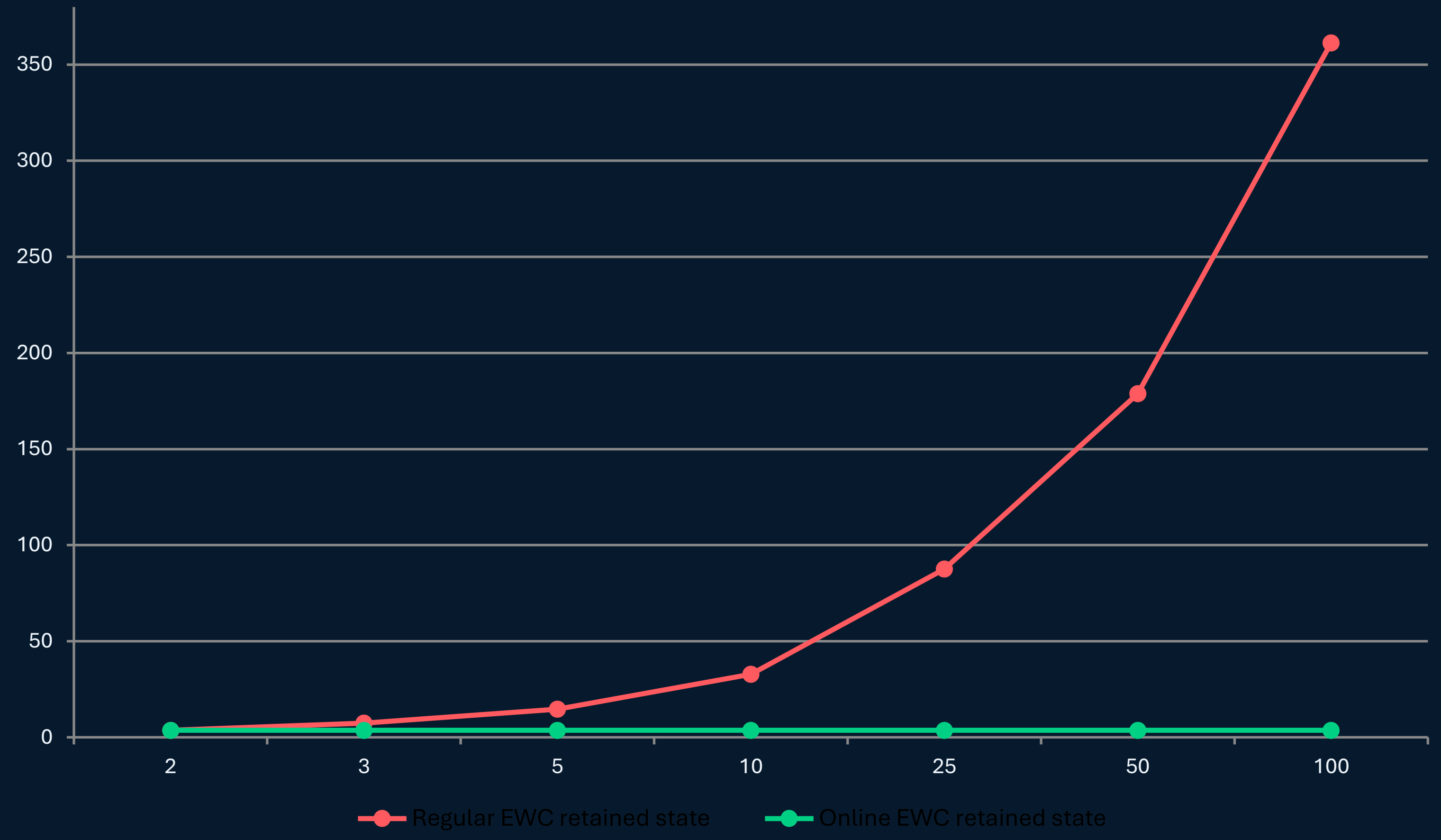
Online EWC state = 2 x parameters x 4 bytes

The model weights are the same in both cases; only the retained EWC history changes.

Scenario	Regular EWC	Online EWC	State saving
3 tasks	7.30 MiB	3.65 MiB	50.0%
10 tasks	32.86 MiB	3.65 MiB	88.9%
100 tasks	361.39 MiB	3.65 MiB	99.0%

Memory: regular EWC grows linearly; Online EWC stays flat

For the same Figure A MLP, the retained EWC state becomes dramatically different as the task sequence grows.



At 100 tasks

361.4 MiB

Regular EWC retained state

3.65 MiB

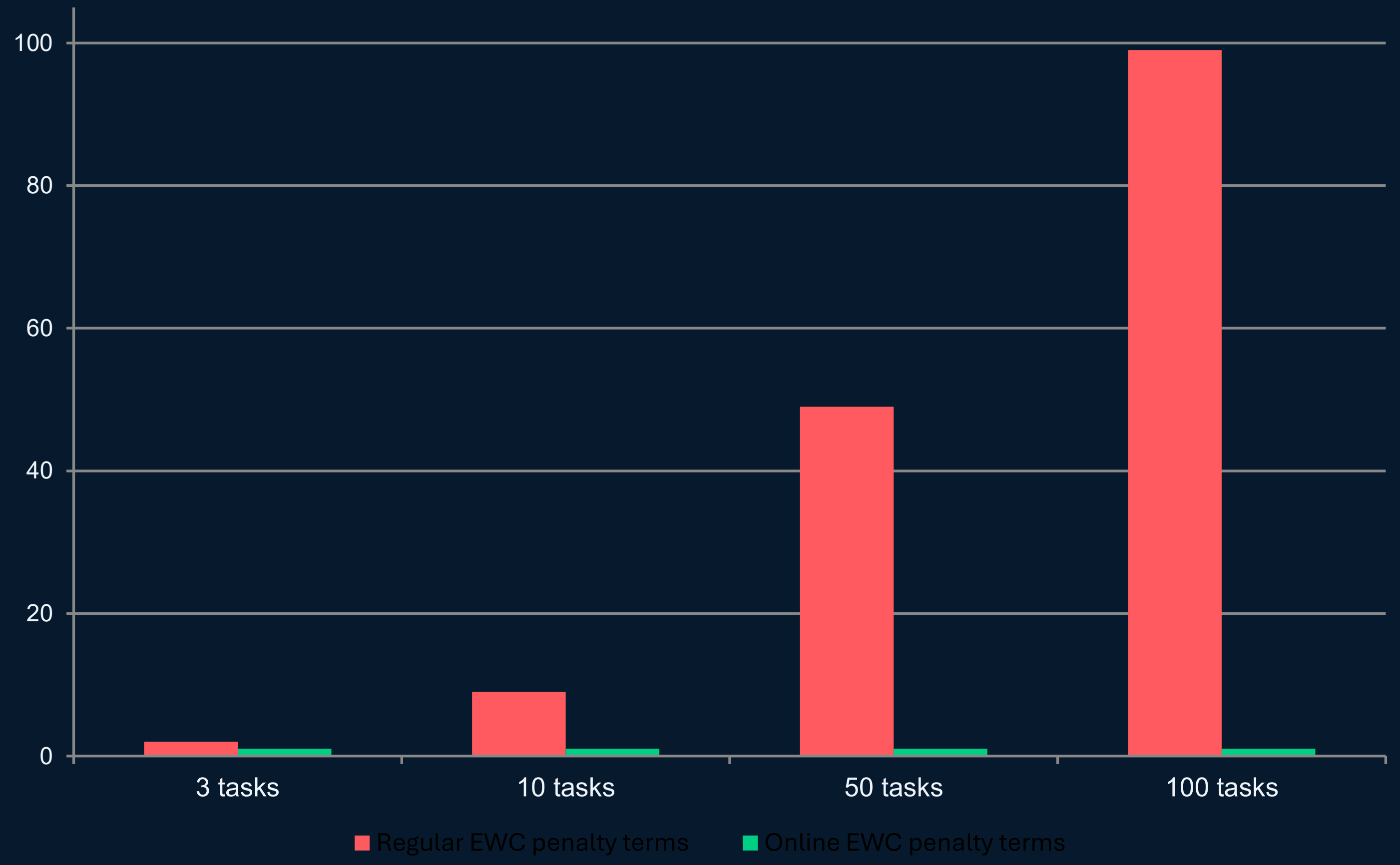
Online EWC retained state

99.0% less retained EWC state

Online EWC savings model - Figure A MLP: 478,410 parameters, float32

Penalty computation: Online EWC keeps one penalty term

During the last task, regular EWC compares against every past task; Online EWC compares against one consolidated state.



2x

lighter penalty calculation in Graph A final task

9x

lighter penalty calculation at 10 tasks

99x

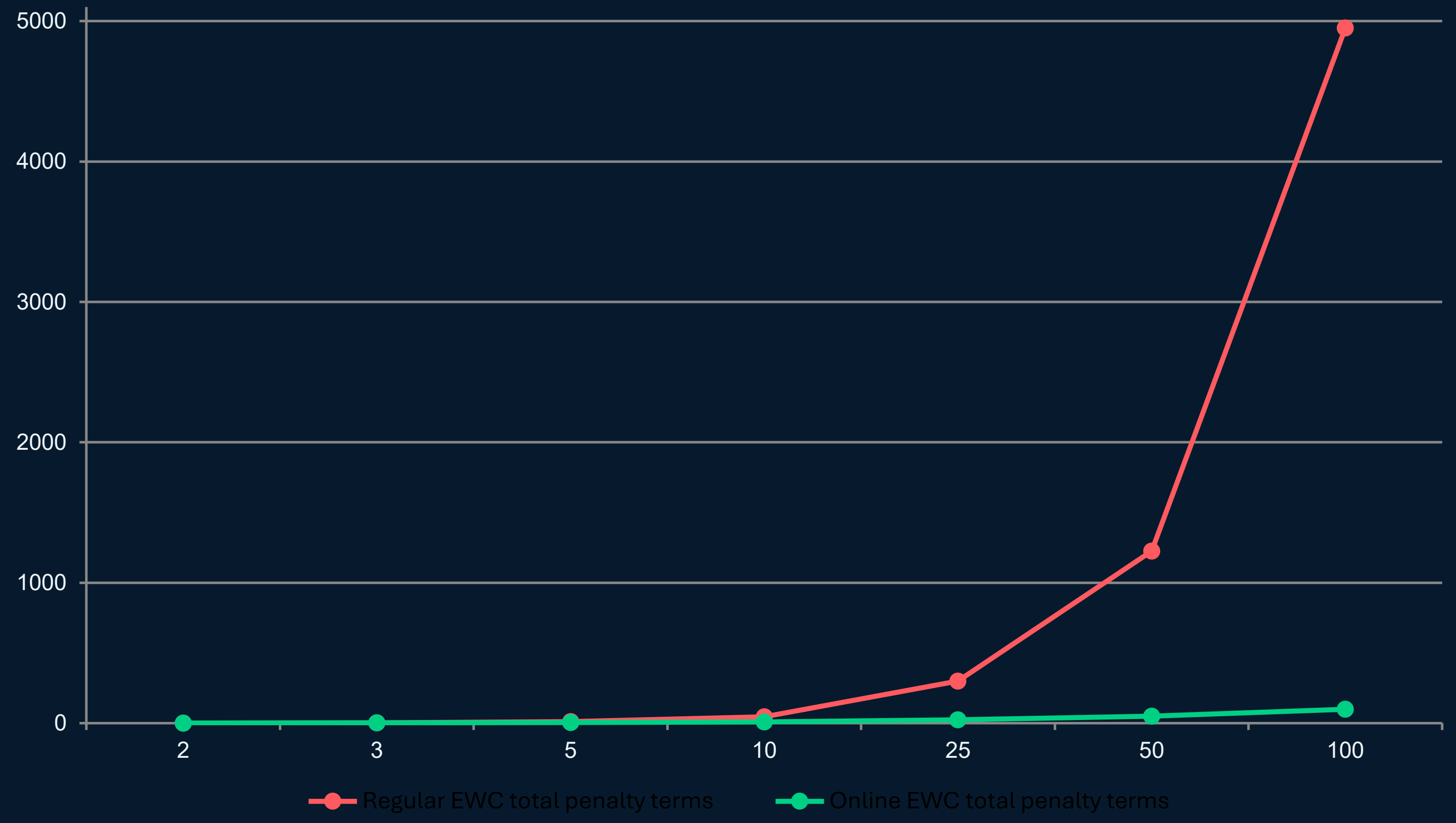
lighter penalty calculation at 100 tasks

Runtime impact in Graph A is modest because most time is still normal training. The scalability win appears as the task sequence grows.

Online EWC savings model - Figure A MLP: 478,410 parameters, float32

Across a full sequence, the penalty workload gap compounds

Total penalty terms accumulated during sequential training: regular grows quadratically; Online grows linearly.



Formula

Regular EWC total = T(T - 1) / 2
Online EWC total = T - 1
Reduction ratio = T / 2

5x

fewer terms over 10 tasks

50x

fewer terms over 100 tasks

Recommended framing for the presentation

Precise, defensible wording for explaining the Graph A implementation.

This implementation is valid as an Online / Consolidated EWC variant.

It preserves the core EWC idea - protect important parameters using Fisher information - but stores one consolidated Fisher + parameter snapshot instead of one pair per previous task.

Graph A

small absolute memory saving, clear conceptual saving

10+ tasks

the difference becomes visibly significant

100 tasks

constant memory becomes the main advantage

One-line takeaway: Online EWC converts EWC history from a growing archive into a single consolidated memory state.